Unit: Unit 5: Linear Equations **Name:** ________________

Topic: Scatter Plot and The Best Fit Line **Date:** __________

Advanced (Carolina Reaper)

Q1. What is the purpose of a scatter plot?
- (A) To show a line of best fit
- (B) To display categorical data
- (C) To visualize the relationship between two variables
- (D) To create a bar graph
- (E) To make predictions without data

Q2. Which of the following correlations is represented by a line with a positive slope?
- (A) Negative correlation
- (B) No correlation
- (C) Positive correlation
- (D) Inverse correlation
- (E) Direct correlation

Q3. What is the equation of a line of best fit for the data points $(2, 3)$, $(4, 5)$, and $(6, 7)$?
- (A) $y = x + 1$
- (B) $y = x - 1$
- (C) $y = 2x - 1$
- (D) $y = 2x + 1$
- (E) $y = x^2$

Q4. If a scatter plot shows a strong positive correlation, what can be said about the line of best fit?
- (A) It has a negative slope
- (B) It has a positive slope
- (C) It is a horizontal line
- (D) It is a vertical line
- (E) It has a slope of 1

Q5. Which of the following is a characteristic of a line of best fit?
- (A) It passes through all data points
- (B) It is a vertical line
- (C) It minimizes the sum of the squared errors
- (D) It has a slope of 0
- (E) It is a curve

Q6. What is the purpose of finding the line of best fit for a set of data?
- (A) To find the equation of the line that passes through all data points
- (B) To determine the correlation between two variables
- (C) To make predictions about future data points
- (D) To visualize the data
- (E) To find the mean of the data

Q7. If a scatter plot shows a weak negative correlation, what can be said about the line of best fit?
- (A) It has a positive slope
- (B) It has a negative slope
- (C) It is a horizontal line
- (D) It is a vertical line
- (E) It has a slope of 0

Q8. Which of the following is an example of making a prediction based on a line of best fit?
- (A) Using the line of best fit to find the equation of the line that passes through all data points
- (B) Using the line of best fit to determine the correlation between two variables
- (C) Using the line of best fit to estimate the value of y for a given value of x
- (D) Using the line of best fit to visualize the data
- (E) Using the line of best fit to find the mean of the data

Open Ended Questions (FRQ)

Q9. Create a scatter plot for the data points $(1, 2)$, $(3, 4)$, $(5, 6)$, and $(7, 8)$. Draw the line of best fit and use it to estimate the value of y for $x = 10$. Show your work and explain your reasoning.

Q10. A bakery sells a certain type of bread. The number of loaves sold per day and the price per loaf are recorded for 5 days: (2,1.50),(4, 1.75), (6,2.00),(8, 2.25), and (10,2.50)$.Createascatterplotofthedataanddrawthelineofbestfit.Usethelineofbestfittopredict$

Chef's Tips

- Tip 1: Ensure students understand the concept of correlation and how it relates to the line of best fit.
- Tip 2: Emphasize the importance of showing work and explaining reasoning in free-response questions.
- Tip 3: Consider providing additional support or scaffolding for students who struggle with creating scatter plots or drawing lines of best fit.